# Intro to Cybersecurity:

Understanding the Basics

# Agenda

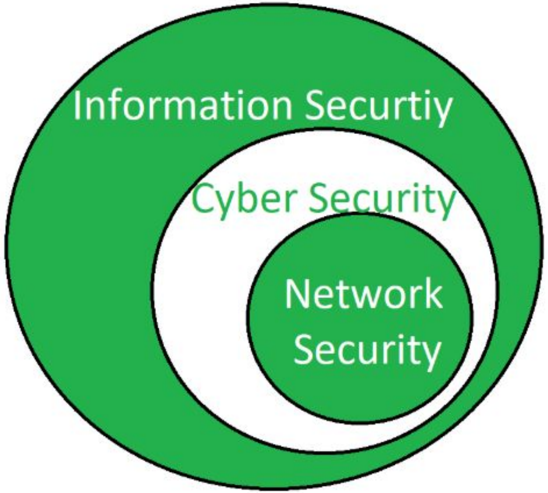
1. Understanding Cybersecurity
2. Common Threats
3. Live Phishing Simulation
4. Good Practices
5. Protecting your Devices
6. Careers/Resources

# What is CYBERSECURITY?

IT Security > **Cybersecurity** > Network Sec

CIA Triad:

- Confidentiality, Integrity, Availability
  - Confidentiality: Protecting data from unauthorized access.
  - Integrity: Ensuring data is accurate, reliable and not to be messed with.
  - Availability: Ensuring data and systems are available and accessible when needed.



# Examples

**Confidentiality**

- Not everyone should have access to your bank account.
- The creation of shared accounts can break the confidentiality scheme

**Integrity**

- Free from modification.
- Data transmitted and received should mirror the same format.
- If you deposit $1000, your account should increase by $1000, not $10000

**Availability**

- Key areas: uptime, storage access, or accessing social media sites.
- Logging on to a system during specific periods.

# Common Terms

**Vulnerability**

Weakness or loophole

Ex: Pw requires 15 chars but a user can create a pw with 4 chars

**Threat**

Dangers

Ex: Malware, phishing emails, unauthorized access attempts

**Risk**

Threat X Vulnerabilities = Risk

Ex: Chance of a data breach (threat) due to a weak password (vulnerability).

**Exploit**

Attacks

Ex: Spreading malware to compromise a network.

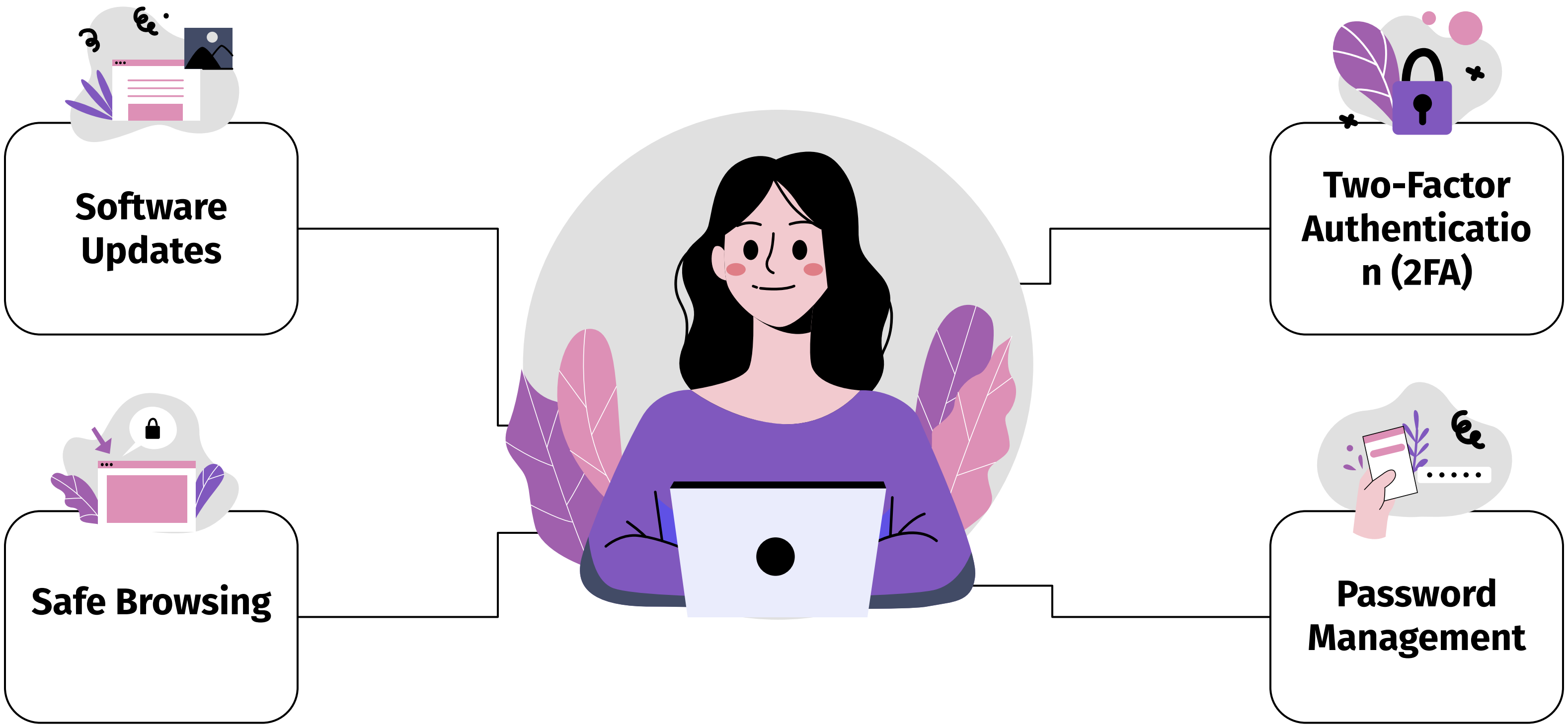
# Common Cyber Threats

| | |
|---|---|
| **Malware (Malicious Software)** | Viruses, Trojans, and ransomware are types of malware that can infect and damage your computer |
| **Phishing** | Scam emails that impersonate trusted entities, aiming to steal personal info |
| **Social Engineering** | Manipulating people. A hacker posing as a co-worker to trick an employee into revealing sensitive data |
| **DDoS (Distributed Denial of Service)** | Overloading a website with traffic to make it inaccessible to users |

# Live Phishing Simulation

https://phishingquiz.withgoogle.com/

# Good Practices



# Software Updates

Explanation: Keeping your software, operating systems, and applications up to date is essential because updates often include security patches that address vulnerabilities. Failing to update leaves your systems at risk of exploitation by cybercriminals.

Ex: When you receive a notification for a system update, like "iOS update available," this update not only adds new features but also fixes known security flaws. By updating, you ensure that it is better protected against potential threats.

# 2-FA

Explanation: 2FA adds an extra layer of security to your accounts. It requires you to provide two different forms of verification before gaining access. This can be something you know (like a password) and something you have (like a temporary code from an authenticator app).

Ex: When you enable 2FA for your email account, after entering your password, you receive a unique code on your mobile app. Only by entering this code can you access your email. Even if someone knows your password, they can't access your account without the second factor.

# Safe Browsing

Explanation: To avoid threats like phishing and malware, be cautious of suspicious links, don't download attachments from unknown sources, and verify the legitimacy of websites and emails.

Ex: Suppose you receive an email with a link claiming to be from your bank, asking you to provide personal information. Instead of clicking the link, visit the bank's official website directly by typing the URL into your browser.

# Password Management

Explanation: Creating strong, unique passwords for each of your online accounts is essential. A password manager can help generate, store, and autofill these complex passwords, making it convenient and secure.

Ex: Suppose you use a password manager to create and store a strong, unique password for your social media account. The password might look like "r$2E#9GfP4q7Zs!". The password manager remembers this for you, so you don't have to write it down or remember it.

# Additional Precautions

**Antivirus software** — Scans/removes malware

Block unauthorized network access — **Firewall**

**Updates** — Very important to keep system up to date

Prevents data loss. Backup to cloud storage — **Backups**

# Check Your Device!

**Windows:**
Settings > Privacy and Security > Windows Security > Check Protection Areas

**MacOS:**
System Preferences > Security and Privacy > Check Protection Areas (Firewall) (Privacy)

# Potential Careers

01

## Pen-Testers

“Good Hackers”
(Ethical Hackers)
Identify system vulnerabilities

02

## Administrators

Manage security systems, networks and policies

03

## Developers

Create secure software focusing on vulnerability prevention

04

## Client-Service Professionals

Assist clients with cyber solutions

# Certifications

**CISSP**

Info sec, risk management

**CompTIA (so many!)**

Net+, Sec+, CySa+, etc.

**CCNA**

Networking concepts

**GIAC (so many!)**

GCIA, GCEC, GPEN

**OSCP**

Pentesting/hacking.
24 hr exam - very difficult

# Resources

Websites:

- Cybrary
- CyberSeek

Books:

- Hacking: The Art of Exploitation, Jon Erickson
- The Cybersecurity Mindset, Dewayne Hart (reading this currently)

Practice:

- HackTheBox
- CyberStart

# Thanks for Attending!